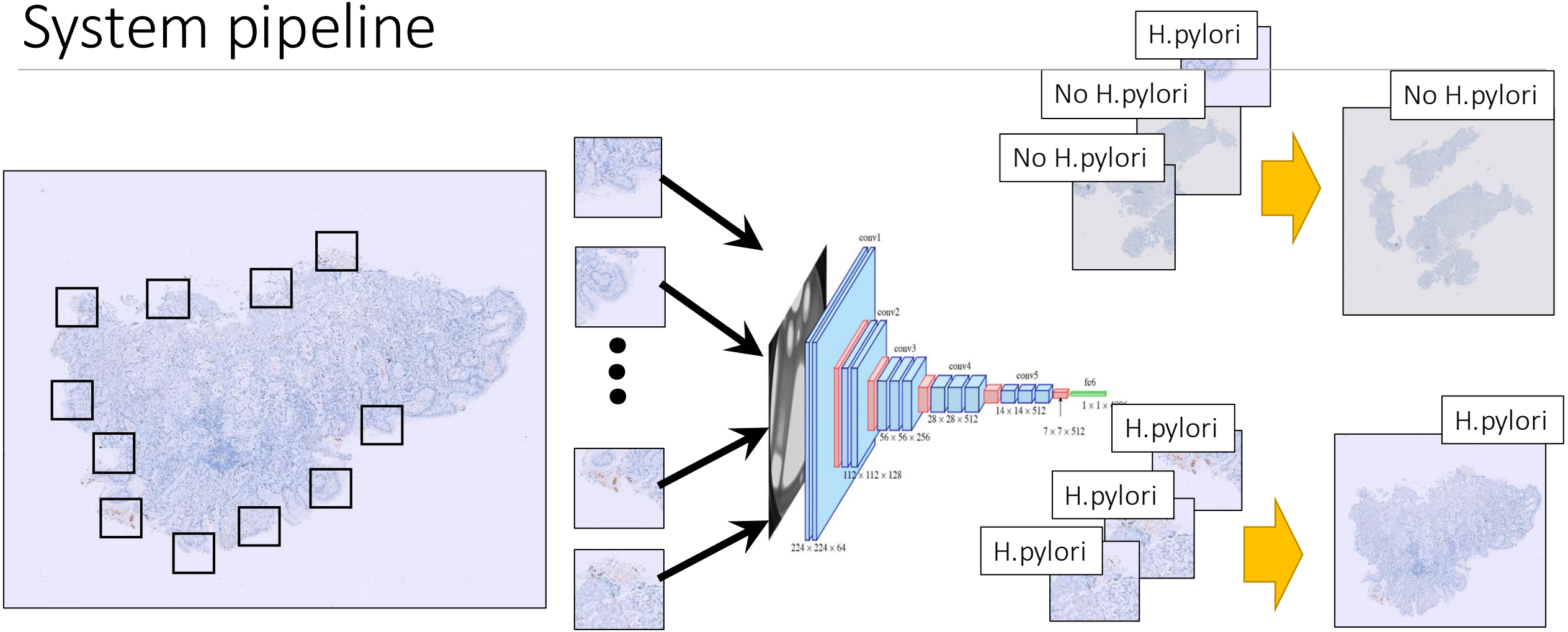
Patch classification

System pipeline



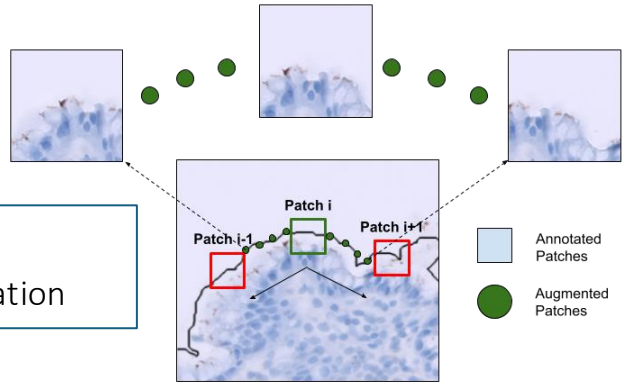
Data preparation:
Extract patches from WSI along sample border

Patch classification:
Identify presence of H.pylori in each patch

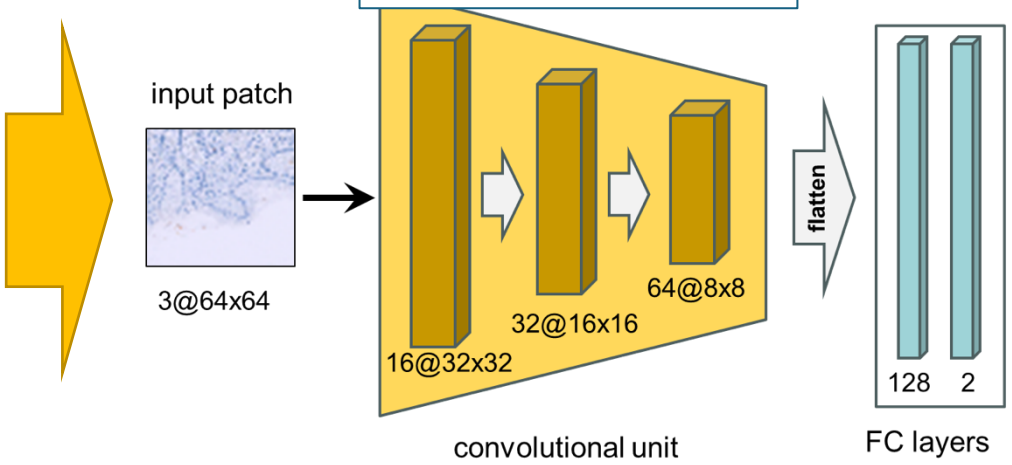
Patient diagnosis:
Aggregate patch output for diagnosis

Patch Classification

Data Augmentation

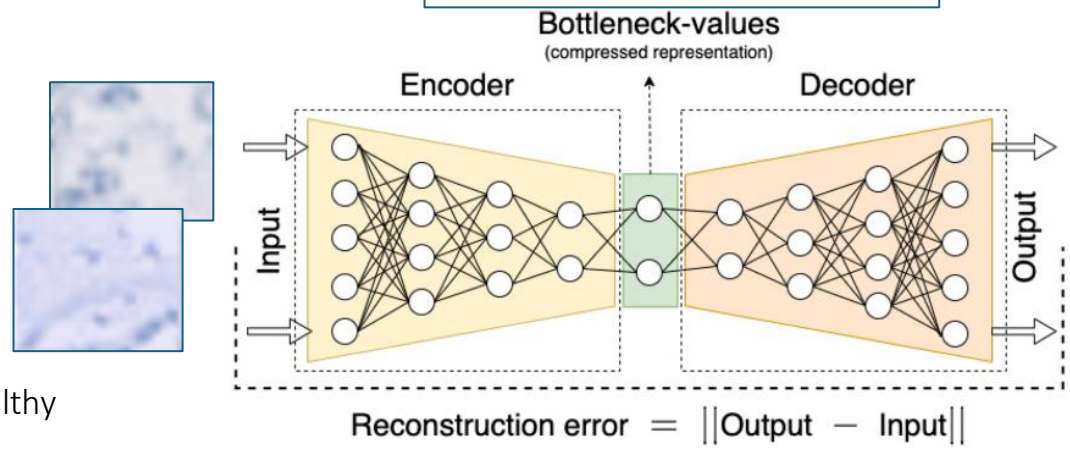


Shallow architecture



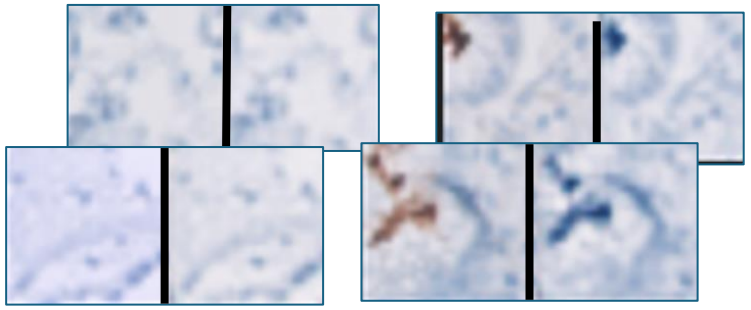
Classification Approach

Self-supervised learning of visual representations.



Healthy

Reconstruction error detects anomalous staining

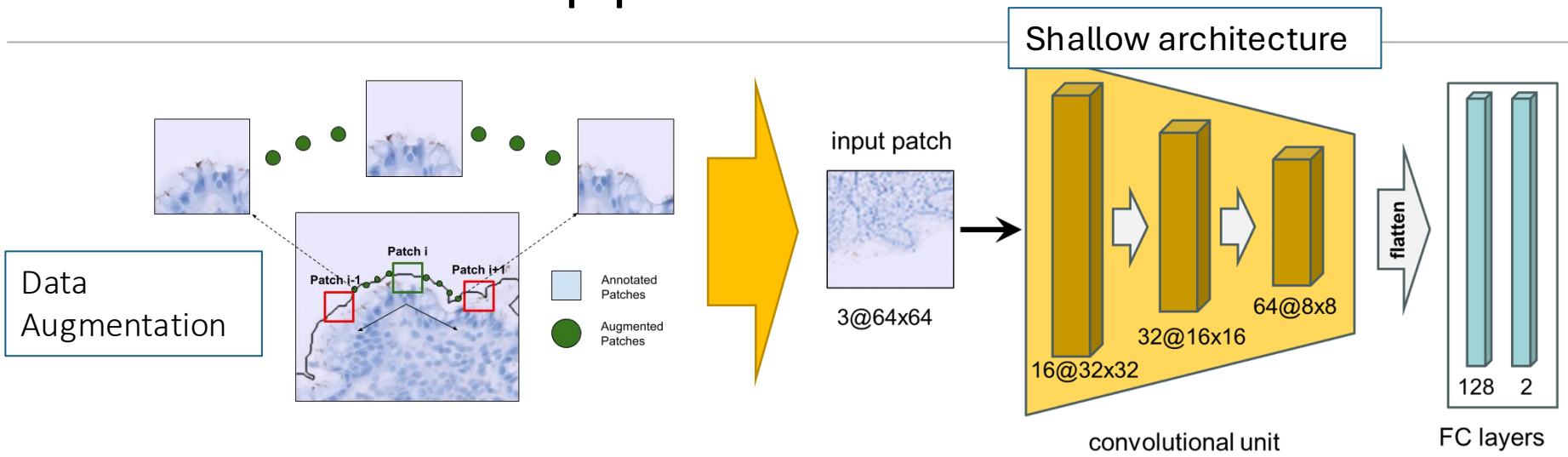


Healthy

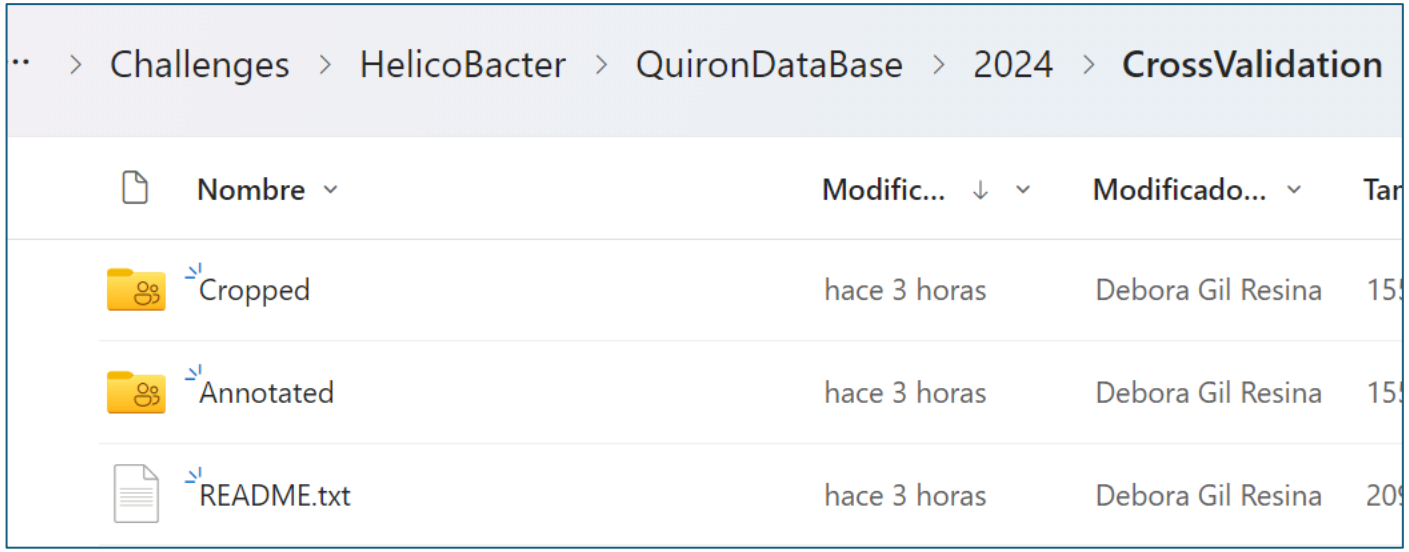
Disease present

Anomaly Detector

Classification Approach



Own Architecture



Cross-validation using the set of Annotated patches

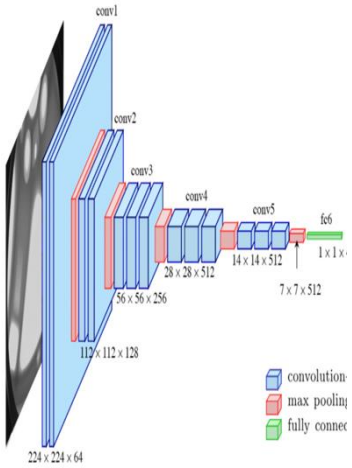
Weighted loss to account for class imbalance

$$loss = \sum_{i=1}^{NClass} w_i loss_i$$

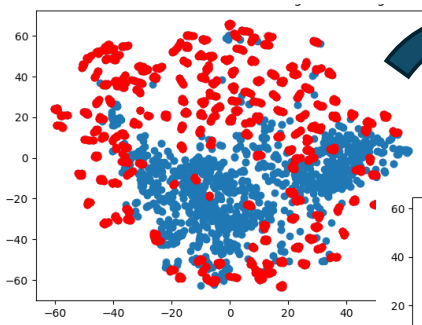
$$w_i = \frac{\#Samples}{\#Class_i}$$

Classification Approach

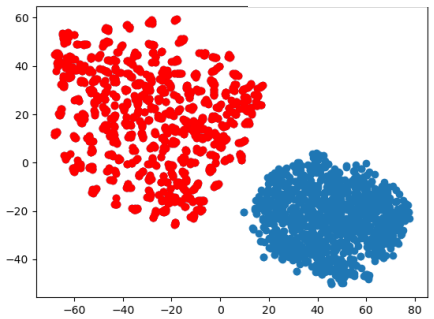
Backbone feature extraction



TSNE visualization (ResNet)



Triplet loss



```
class FeatureExtractorResnet152(nn.Module):
    def __init__(self, model_name, cut_index=None):
        super().__init__()

       self.fe  = models.resnet152(pretrained = True)
       self.fe.fc = nn.Identity()

    def forward(self, x):
        x = self.fe(x)
        return x
```

Pre-Trained Architecture

Contrastive representation learning

$$\mathcal{L}(\mathbf{q}, \mathbf{k}^+, \mathbf{k}^-) = \max(0, D(\mathbf{q}, \mathbf{k}^+)^2 - D(\mathbf{q}, \mathbf{k}^-)^2 + m)$$

q,k+ are from the same class

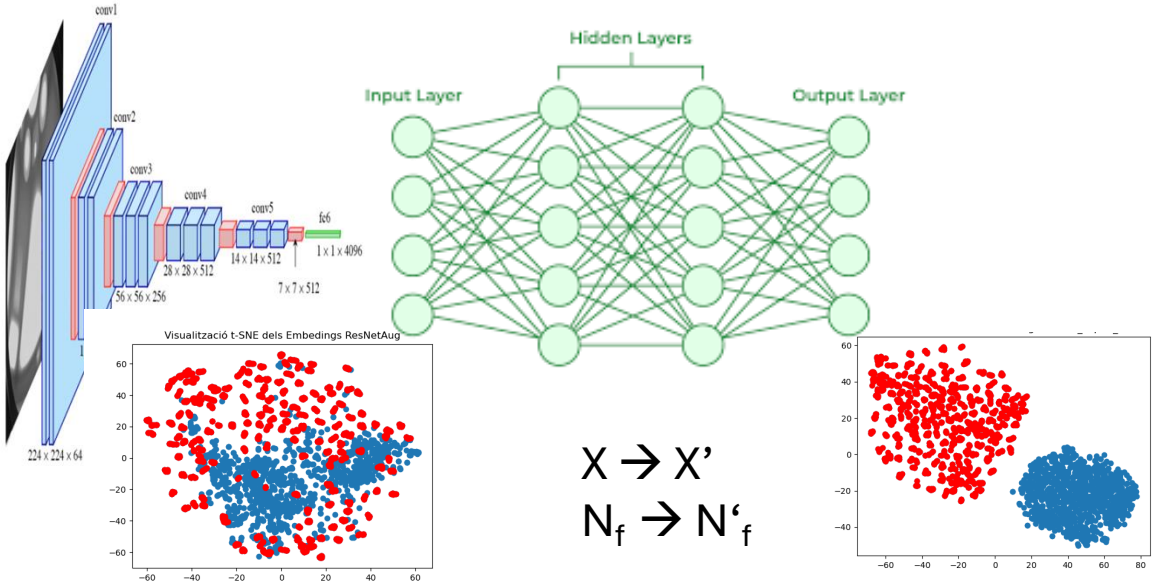
m is the margin between positive and negative classes (mínimum is achieved when $D^+ + m = D^-$)

Classification Approach

Pre-Trained Architecture

Two FC networks are trained:

1. Contrastive.



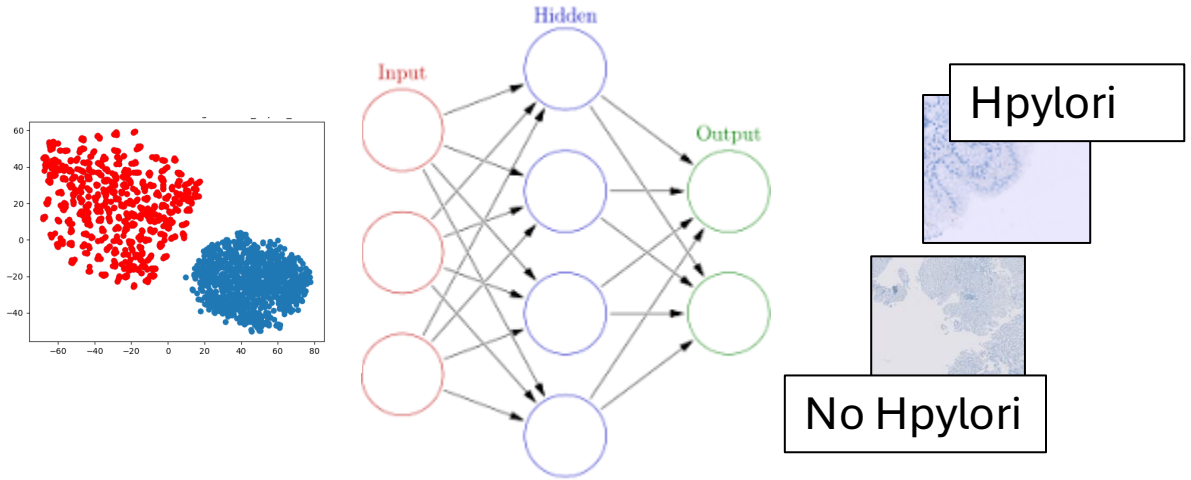
$X \rightarrow X'$

$N_f \rightarrow N'_f$

Transformation of pre-trained space (N_f dim) into more discriminative one (N'_f dim)

Contrastive loss

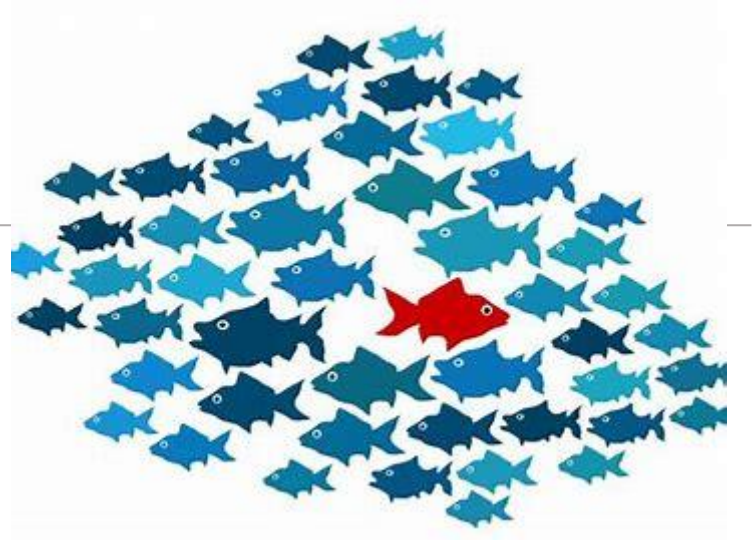
2. Classification

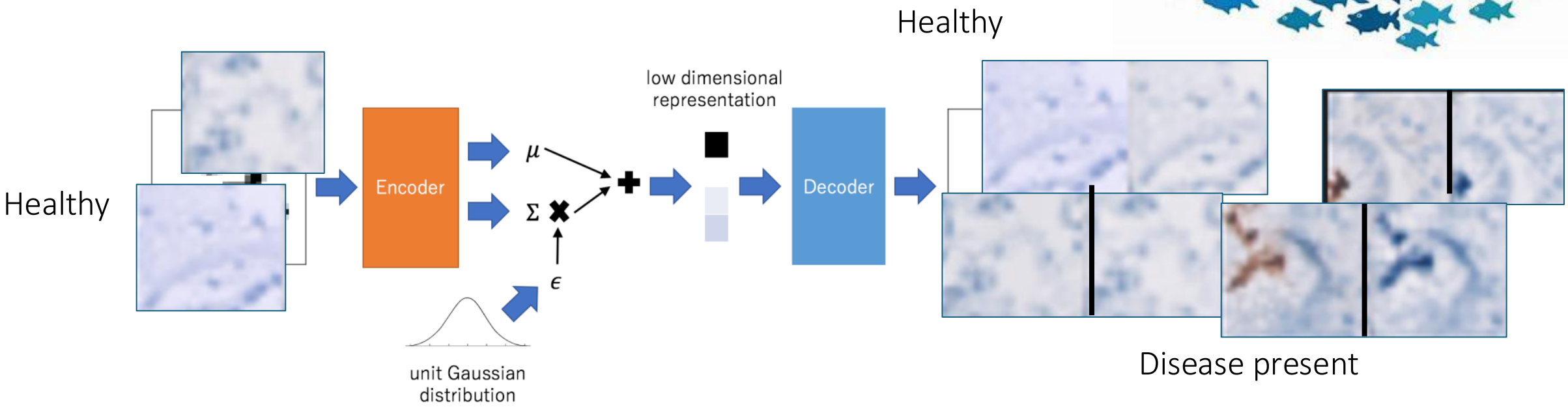


Contrastive space is input to binary classifier for patch classifcation

Weighted cross-entropy

Anomaly Detection

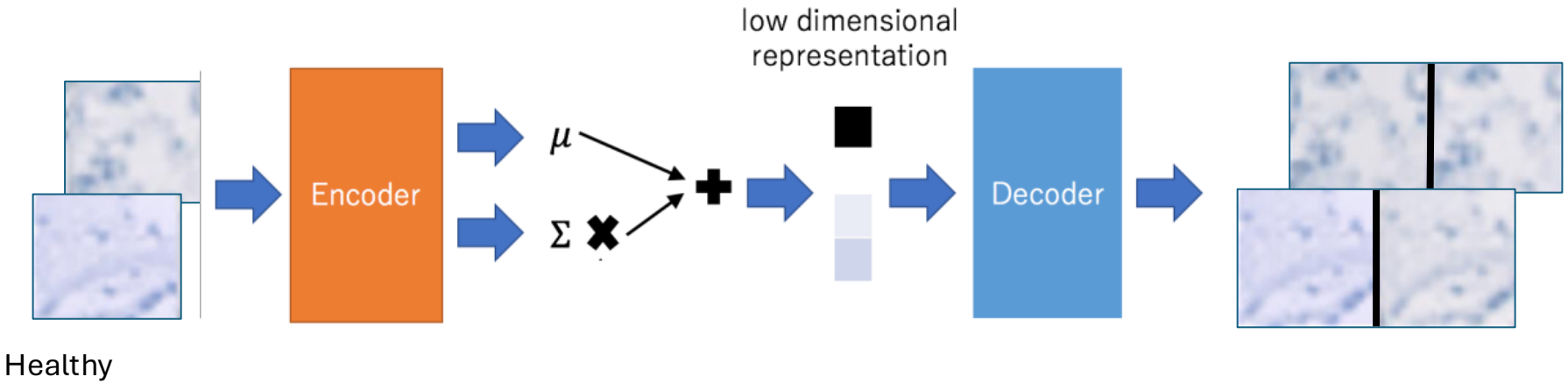




Learn majority/normality (autoencoders). Use reconstruction error to detect the rare small sample size population (https://arxiv.org/abs/2309.16053)

Use generative models for self-supervised learning of representation spaces

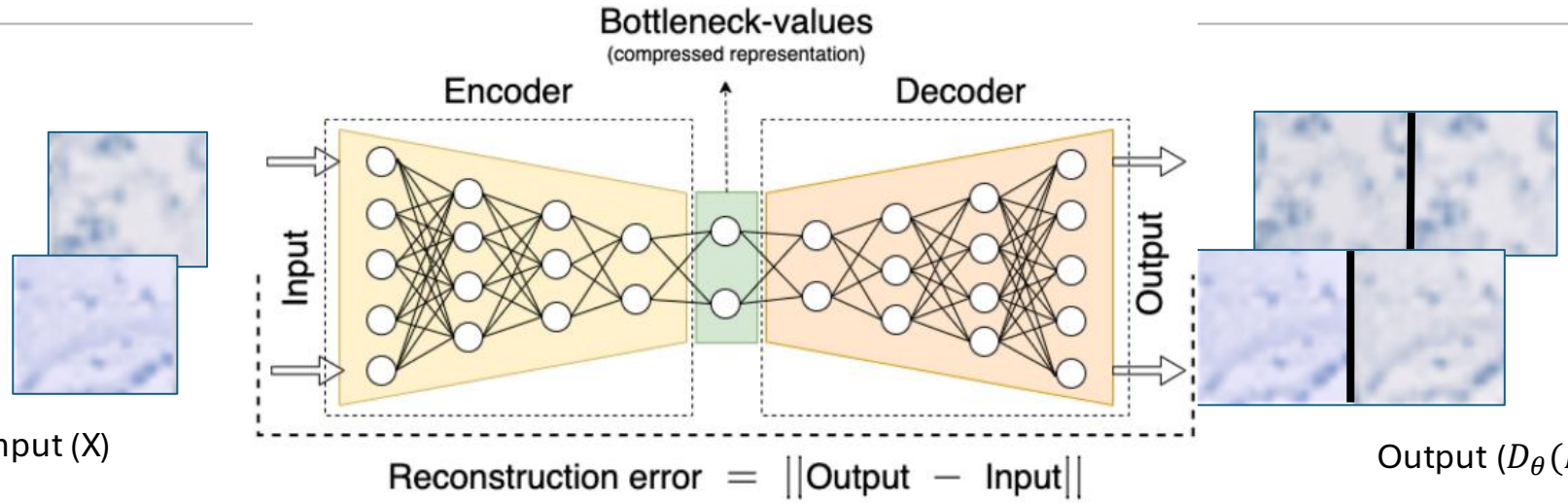
Anomaly Detection



Own Architecture

- Autoencoder trained using patches from healthy patients of the CrossValidation set
- A shallow architecture can be used

Anomaly Detection



Input (X)

Output ($D_\theta(E_\phi(X))$)

Loss: Similarity measure, reconstruction error:

$$L(\theta, \phi) = \frac{1}{N} \sum_{i=1}^{N} \|x_i - D_\theta(E_\phi(x_i))\|_2^2$$

AE components:

Encoder: CNN architecture with dim reduction (maxpool) to extract visual representation space (latent space)

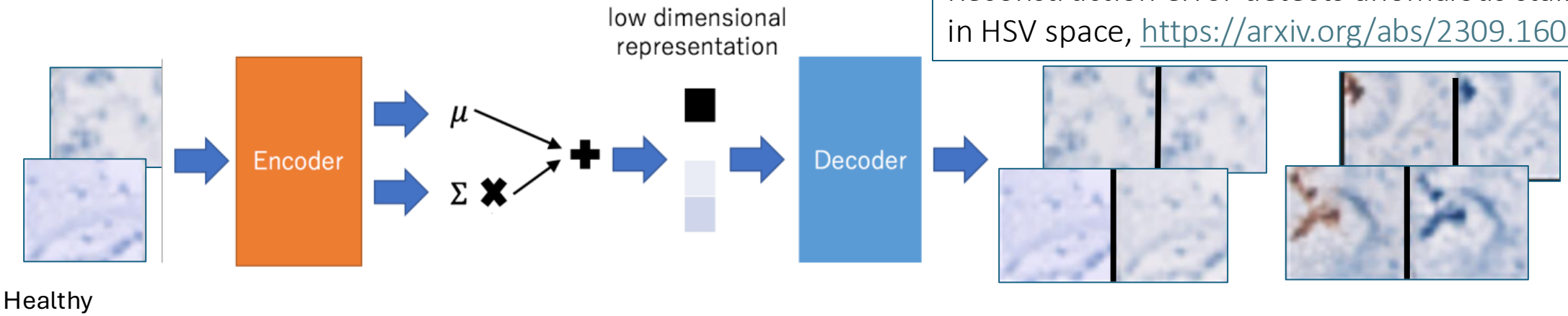
Bottleneck: block to reduce the dimensionality and obtain a compact representation (Z)

Decoder: CNN architecture with upsampling to restore original content from latent space

Anomaly Detection

$$F_{red} = \frac{\#\{(i,j) \text{ with } -20 < H_{Ori}(i,j) < 20\}}{\#\{(i,j) \text{ with } -20 < H_{Rec}(i,j) < 20\}}$$

Reconstruction error detects anomalous staining (red channel in HSV space, https://arxiv.org/abs/2309.16053).

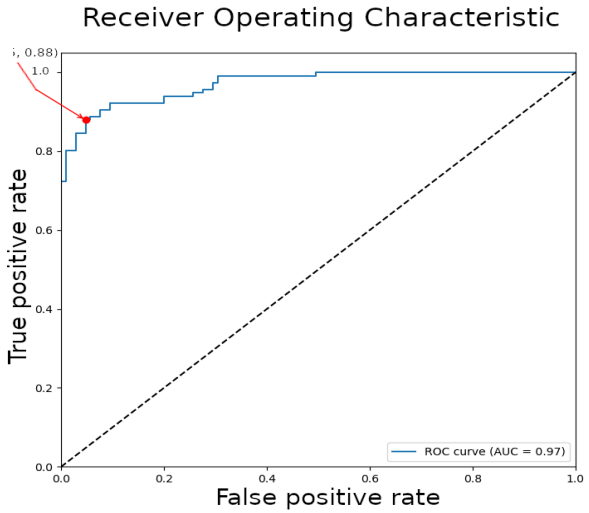


Threshold on the error function detects Hpilory in patch

Optimal value learned from Annotated patches using ROC adaptive thresholding

Error > Th → Hpilory

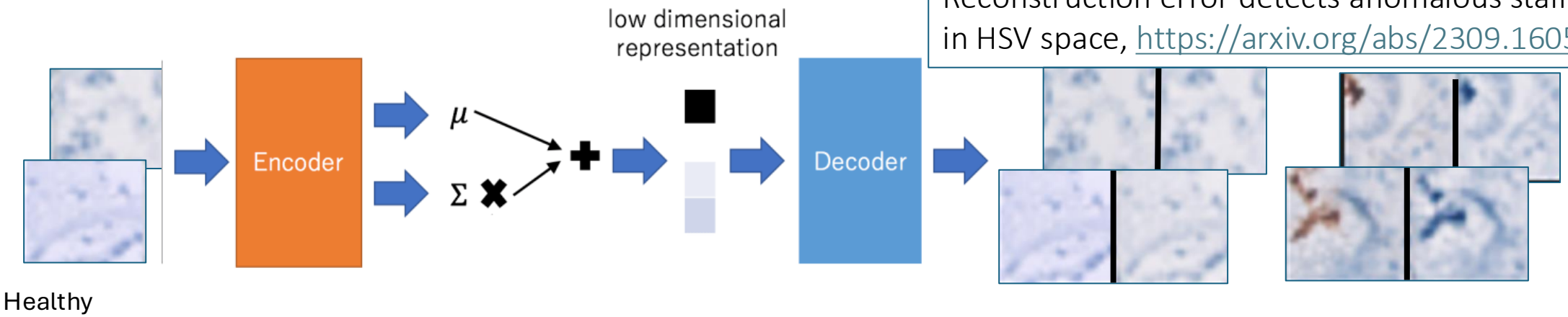
Optimal th



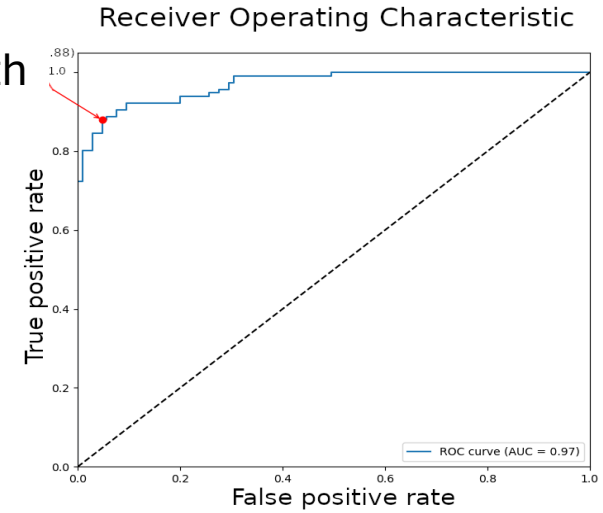
Anomaly Detection

$$F_{red} = \frac{\#\{(i,j) \text{ with } -20 < H_{Ori}(i,j) < 20\}}{\#\{(i,j) \text{ with } -20 < H_{Rec}(i,j) < 20\}}$$

Reconstruction error detects anomalous staining (red channel in HSV space, https://arxiv.org/abs/2309.16053).



Optimal th



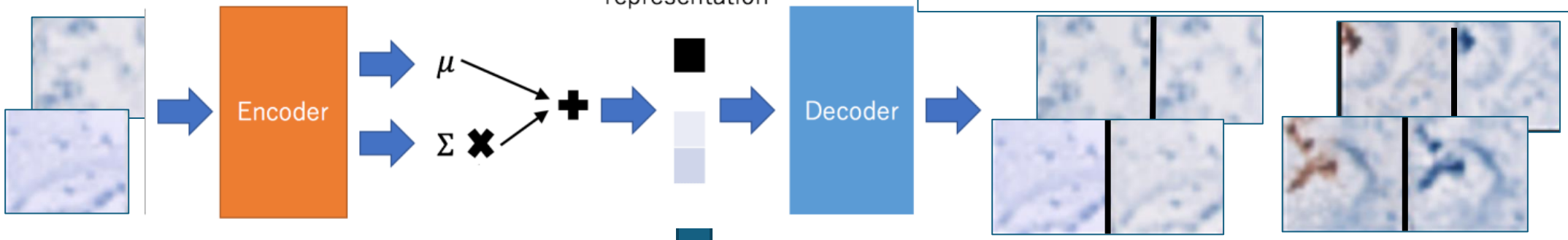
ROC shows compromise between the precision and recall obtained by increasing Th

Optimal Th is closest point to (0,1) representing FPR=0 with TPR=100%

AUC (Area under the curve) is also a measure of the system performance

Anomaly Detection

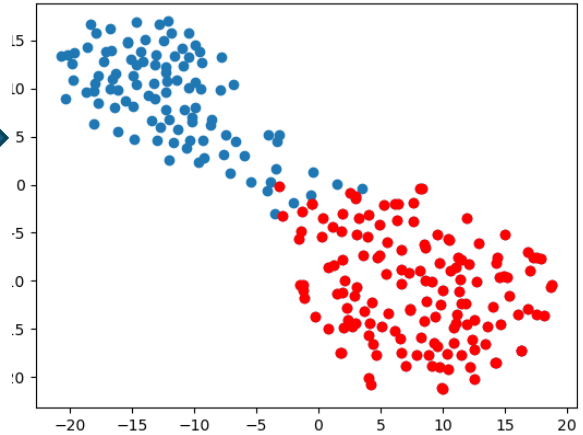
Reconstruction error detects anomalous staining (red channel in HSV space, https://arxiv.org/abs/2309.16053).



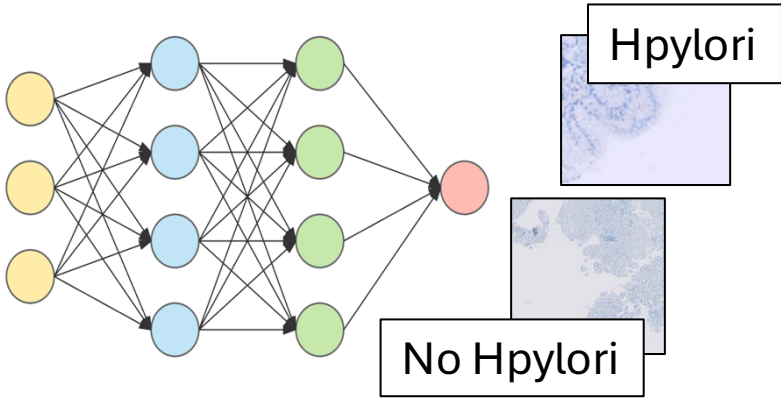
Healthy

Self-supervised learning of visual representations.

Alternatively.....



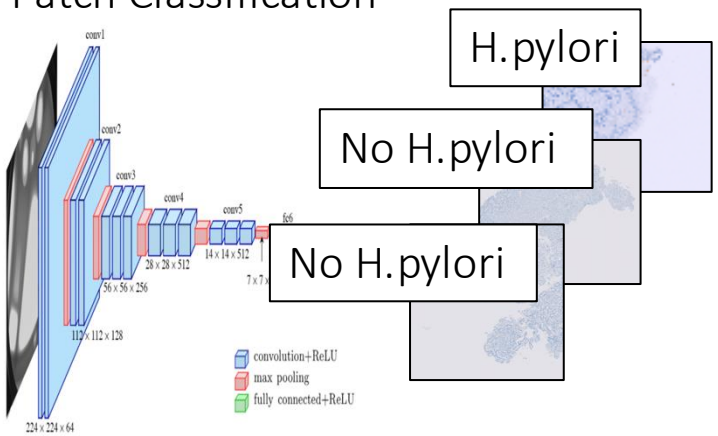
Own Architecture



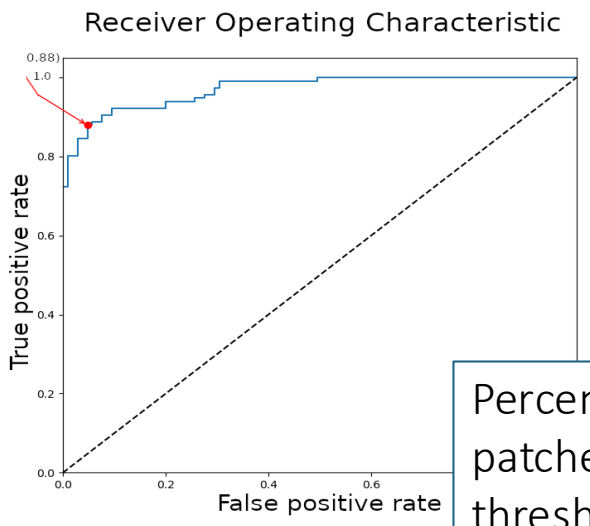
Patient Diagnosis

Adaptive Thresholding

Patch Classification

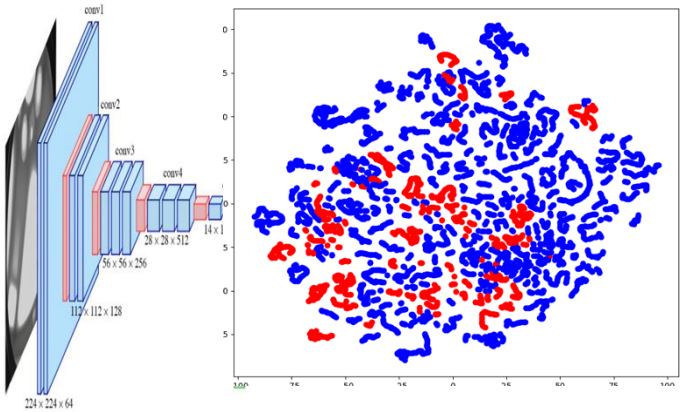


Optimal th

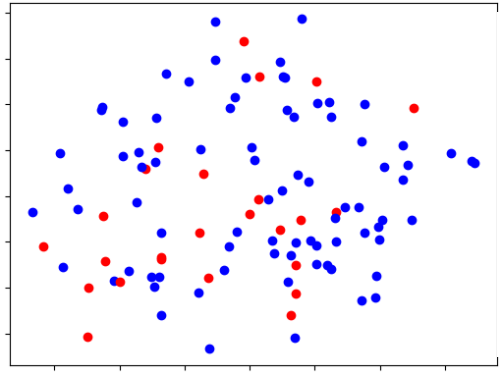


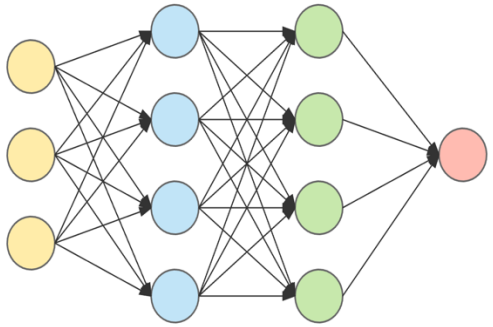
Percentage of positive patches with adaptative thresholding

Attention Mechanism

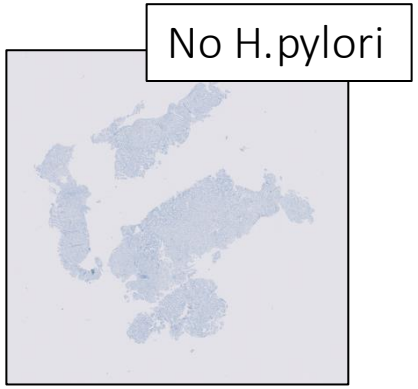


Patch Visual Features





Sample representation space using attention for patient classification



Sample Classification